

Internship Report – Day 2

Date: 06/05/2025

Name: C H UDAY KUMAR

Roll.no: BU22CSEN0300538

Day2 Overview:

On the second day of the internship, we were introduced to Automatic Number Plate Recognition (ANPR) systems. We were assigned to watch an educational YouTube video on how ANPR works using YOLOv8 and EasyOCR, and to understand the step-by-step implementation. The video explained the use of object detection and OCR techniques to extract and recognize license plate numbers from video frames. It also covered challenges like varying formats and preprocessing steps to improve accuracy. Along with this, we were instructed to provide a relevant number plate dataset link for future model training.

Tools Used Today:

YouTube – For watching the ANPR tutorial video.

Kaggle – For exploring and sharing a number plate dataset.

ChatGPT – For summarizing the video, understanding the concept, and generating the report.

Task Given:

Watch a YouTube video on ANPR (Automatic Number Plate Recognition), share the video link, write a brief summary, and provide a dataset link related to number plates

Approach to the Task:

YOUTUBE LINK: <https://youtu.be/fyJB1t0o0ms?si=9RslwiQqp260xduH>

KAGGLE LINK: <https://www.kaggle.com/datasets/saisirishan/indian-vehicle-dataset>

Overview of Automatic Number Plate Recognition (ANPR)

- The tutorial demonstrates building an ANPR system using Python, YOLOv8 for object detection, and EasyOCR for text recognition.
- The primary goals include detecting license plates from video frames and reading the text contained within those plates.
- The tutorial showcases a practical application, utilizing a video of vehicles on a highway as the data source.

Key Technologies Used

- YOLOv8: An object detection framework that efficiently identifies and locates license plates in video frames.
- EasyOCR: A Python library utilized to read and extract text from the detected license plates.
- SORT Algorithm: Implements object tracking to maintain the association between detected vehicles and their respective license plates across frames.

License Plate Format and Challenges

- The tutorial focuses on the UK license plate format, which consists of seven characters: two letters, two numbers, followed by three letters (e.g., AB12CDE).
- The diversity in global license plate formats presents challenges in creating a universally applicable ANPR system.
- The tutorial simplifies the problem by concentrating on a specific format, making the implementation more manageable.

Data Preparation and Implementation Steps

- The dataset used for training the license plate detector is referenced, and a link to this dataset is provided in the video description.
- The coding process is structured into a pipeline, starting with model loading, video capture, and frame processing.
- Functions for reading license plates and managing vehicle detection (get car function) are highlighted but initially left unimplemented.

Image Processing Techniques

- Before text extraction, the license plate images undergo preprocessing, which includes converting to grayscale and applying thresholding to enhance readability for OCR.

- The preprocessing step ensures the images are in a suitable format for EasyOCR, improving recognition accuracy.

Result Compilation and Visualization

- Detected license plate data is organized into a CSV file, storing information such as frame number, car ID, bounding boxes, and the detected text with corresponding confidence scores.
- The tutorial discusses handling missing frames and emphasizes the importance of correctly associating license plates with vehicles for accurate data representation.
- Visualization scripts are mentioned, allowing users to create a video that overlays detected license plate information on the original footage.

Future Improvements and Considerations

- The tutorial concludes by suggesting further enhancements, such as refining the license plate reading function and improving visualization stability through interpolation of missing data.
- The importance of confidence scoring in determining the most reliable text detection across multiple frames is also discussed, emphasizing the need for robust data handling in ANPR systems.